

# Industrial Electrical System Questions & Answers

## Detailed Solutions

### Theory & Application Questions

3. Justify whether you recommend a Switch Fuse Unit (SFU) or a MCB for giving power to a three-phase 2 kW Induction motor. Pros and cons of your recommendation should be brought clearly.

**Recommendation:** I would recommend a **Miniature Circuit Breaker (MCB)** (specifically a TP-MCB with C-Curve or D-Curve characteristics) for a 2 kW induction motor.

**Justification:** A 2 kW three-phase motor draws approximately 4 Amps of full load current.

$$I = \frac{P}{\sqrt{3} \times V \times \cos \phi} \approx \frac{2000}{1.732 \times 415 \times 0.8} \approx 3.5 \text{ to } 4 \text{ A}$$

For such small ratings, an MCB offers superior protection and operational convenience compared to a rewirable fuse.

#### Pros of MCB Recommendation:

- **Protection against Single Phasing:** If an internal fault occurs, a Three-Phase MCB trips all three poles simultaneously. In a Switch Fuse Unit, one fuse might blow while the others remain intact, causing the motor to run on two phases (single phasing), which burns the winding.
- **Sensitivity:** MCBs are more sensitive to small overloads (e.g., 1.2 times rated current) compared to fuses.
- **Quick Restoration:** In case of a trip, the MCB can be instantly reset without searching for replacement wire.

#### Cons:

- **Cost:** MCBs are more expensive than a simple Switch Fuse Unit.
- **Inrush Tripping:** If the wrong type (e.g., Type B) is selected, the high starting current of the motor might cause nuisance tripping. (Solved by using Type C).

4. A metal cutting workshop is having 110 kW, 3 phase load at 400 V. Recommend with detail justifications whether you will use MCB or MCCB as a main switch before the distribution box.

**Recommendation:** I strongly recommend using an **MCCB (Moulded Case Circuit Breaker)** as the main switch. An MCB is not suitable for this application.

### Detailed Justification:

#### (a) Current Rating Calculation:

$$I_L = \frac{P}{\sqrt{3} \times V_L \times \cos \phi}$$

Assuming a typical power factor ( $\cos \phi$ ) of 0.8:

$$I_L = \frac{110 \times 10^3}{\sqrt{3} \times 400 \times 0.8} = \frac{110,000}{554.25} \approx \mathbf{198.4 \text{ Amperes}}$$

- (b) **Limitation of MCB:** Standard MCBs available in the market have a maximum current rating of **100A or 125A**. Since the load current ( $\approx 198A$ ) exceeds this limit, an MCB cannot be physically used.
  - (c) **Suitability of MCCB:** MCCBs are designed for higher current ratings (typically 100A to 2500A). A 250A MCCB would be appropriate here.
  - (d) **Interrupting Capacity (kA):** An industrial workshop will have a high fault current level (potential short circuit current). MCCBs have a high breaking capacity (e.g., 35kA or 50kA), whereas MCBs are typically limited to 10kA, making the MCCB safer for main incomers.
5. Discuss the constructional features of a MCB clearly indicating the function of each part. Discuss the working principle of RCCB indicating at least three applications of the same.

## A. Constructional Features of MCB

An MCB consists of the following main parts:

- **Bimetallic Strip:** A strip made of two different metals. *Function:* It provides **Overload Protection**. When current exceeds the rated limit for a sustained period, the strip heats up, bends, and unlatches the mechanism to trip the breaker.
- **Electromagnetic Coil (Solenoid):** A coil with a movable plunger. *Function:* It provides **Short Circuit Protection**. During a sudden high fault current, the strong magnetic field instantly pulls the plunger to trip the latch mechanism (Magnetic trip).
- **Arc Chute:** A stack of parallel metal plates. *Function:* To split, cool, and extinguish the electric arc that forms when the contacts separate.
- **Operating Mechanism:** A spring-loaded system. *Function:* Ensures "Quick Make" and "Quick Break" operation independent of the speed at which the user flips the lever.

## B. Working Principle of RCCB

The Residual Current Circuit Breaker (RCCB) works on **Kirchhoff's Current Law (KCL)**.

- It contains a **Core Balance Current Transformer (CBCT)**—a toroidal core with the Phase and Neutral wires passing through it.
- **Normal Condition:** Current flowing in (Phase) = Current flowing out (Neutral). The vector sum is zero, so no magnetic flux is produced in the core.
- **Fault Condition:** If there is leakage to earth, the returning current in the Neutral is less than the Phase current ( $I_L \neq I_N$ ).
- This difference creates a residual flux in the core, inducing a voltage in a secondary sensing coil, which activates a relay to trip the circuit.

#### Applications of RCCB:

- (a) **Domestic Wiring:** To prevent fatal electric shocks to residents in homes.
- (b) **Wet Areas:** Essential in bathrooms and kitchens where water increases shock risk.
- (c) **Fire Protection:** To detect small leakage currents (e.g., 300mA) that can cause insulation heating and fire over time.

#### 6. Discuss main structural parts of an Energy meter.

The induction-type energy meter consists of four distinct systems:

- (a) **Driving System:** Consists of two electromagnets:
  - *Series Magnet (Current Coil):* Connected in series with the load; produces flux proportional to current.
  - *Shunt Magnet (Pressure Coil):* Connected in parallel; produces flux proportional to voltage.

These magnets create a rotating magnetic field.

- (b) **Moving System:** A light aluminum disc mounted on a vertical spindle. It is placed in the air gap between the two electromagnets. The interaction of the fluxes and eddy currents creates a torque that rotates the disc.
- (c) **Braking System:** A permanent magnet (usually C-shaped) placed near the edge of the disc. As the disc rotates, it cuts the magnet's field, inducing eddy currents that oppose the motion (Lenz's Law). This limits the speed of the disc to be proportional to power.
- (d) **Registering System:** A gear train connected to the spindle which drives a series of dials or a cyclometer counter to record the number of revolutions (calibrated in kWh).

#### 7. What is relay? How many types of relays are there? For giving overload protection to an induction motor, which type of relay should be used?

**Definition:** A relay is a sensing device that detects an abnormal condition (fault) in an electrical circuit and sends a command signal to the Circuit Breaker to disconnect the faulty element. It acts as the "brain" of the protection system.

#### Types of Relays (Broad Classification):

- **Based on Construction:**

- (a) Electromagnetic Attraction Type
- (b) Electromagnetic Induction Type
- (c) Thermal Relays
- (d) Static / Solid State Relays
- (e) Microprocessor / Numerical Relays

- **Based on Function:** Over-current, Over-voltage, Differential, Distance, etc.

**For Induction Motor Overload Protection:** The most suitable type is the **Thermal Overload Relay** (specifically the Bimetallic type). It mimics the heating characteristics of the motor windings and trips before the insulation is damaged by heat.

#### 8. Compare the relative merits and demerits of MCB and re-wirable fuse.

| Feature                  | MCB (Miniature Circuit Breaker)                                                 | Re-wirable Fuse (Kit-Kat)                                                                       |
|--------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Sensitivity</b>       | <b>Merit:</b> Very sensitive. Trips even on small overloads (e.g., 20% excess). | <b>Demerit:</b> Low sensitivity. Requires high current to melt; may not detect small overloads. |
| <b>Restoration</b>       | <b>Merit:</b> Easy restoration. Just flip the knob ON.                          | <b>Demerit:</b> Time-consuming. Requires replacing the fuse wire manually.                      |
| <b>Safety</b>            | <b>Merit:</b> Fully enclosed. No risk of accidental touch.                      | <b>Demerit:</b> Risk of shock during wire replacement; sparking may occur.                      |
| <b>Cost</b>              | <b>Demerit:</b> Higher initial cost.                                            | <b>Merit:</b> Very low initial cost.                                                            |
| <b>Calibration</b>       | <b>Merit:</b> Factory calibrated and sealed. Reliable rating.                   | <b>Demerit:</b> Depends on the wire used by the electrician. Wrong wire size ruins protection.  |
| <b>Breaking Capacity</b> | <b>Merit:</b> Better arc quenching capabilities (up to 10kA).                   | <b>Demerit:</b> Low breaking capacity. Can explode under high fault currents.                   |

9. For one domestic consumer, total unit consumed in a month is 299 kWh. Fixed demand charge is Rs 15.00. Meter rent Rs 10.00. Power tariff is as given below:

- **First 25 units:** 518 P/unit
- **Next 35 units:** 569 P/unit
- **Next 40 units:** 670 P/unit
- **Next 50 units:** 745 P/unit
- **Next 50 units:** 762 P/unit
- **Next 100 units:** 762 P/unit
- **Above 300 units:** 762 P/unit

Calculate the gross amount of bill without taxes.

**Solution:**

**Given Data:**

- Total Units Consumed = 299 kWh
- Fixed Demand Charge = Rs. 15.00
- Meter Rent = Rs. 10.00
- Note: Rates are in Paise (P). Conversion: 100 P = Rs. 1.

**Step 1: Energy Charge Calculation (Slab-wise)** We breakdown the 299 units based on the given slabs:

(a) **Slab 1 (0 - 25 units):**

$$25 \text{ units} \times \text{Rs. } 5.18 = \text{Rs. } 129.50$$

$$\text{Remaining units: } 299 - 25 = 274$$

(b) **Slab 2 (Next 35 units):**

$$35 \text{ units} \times \text{Rs. } 5.69 = \text{Rs. } 199.15$$

$$\text{Remaining units: } 274 - 35 = 239$$

(c) **Slab 3 (Next 40 units):**

$$40 \text{ units} \times \text{Rs. } 6.70 = \text{Rs. } 268.00$$

$$\text{Remaining units: } 239 - 40 = 199$$

(d) **Slab 4 (Next 50 units):**

$$50 \text{ units} \times \text{Rs. } 7.45 = \text{Rs. } 372.50$$

$$\text{Remaining units: } 199 - 50 = 149$$

(e) **Slab 5 (Next 50 units):** Since 149 units remain, we utilize this full slab.

$$50 \text{ units} \times \text{Rs. } 7.62 = \text{Rs. } 381.00$$

$$\text{Remaining units: } 149 - 50 = 99$$

- (f) **Slab 6 (Next 100 units):** We have 99 units remaining. This fits entirely in this slab.

$$99 \text{ units} \times \text{Rs. } 7.62 = \text{Rs. } 754.38$$

*Remaining units:* 0

**Step 2: Total Energy Charges**

$$\text{Total Energy Charge} = 129.50 + 199.15 + 268.00 + 372.50 + 381.00 + 754.38$$

$$\text{Total Energy Charge} = \text{Rs. } \mathbf{2,104.53}$$

**Step 3: Gross Amount (Without Taxes)**

$$\text{Gross Amount} = \text{Total Energy Charge} + \text{Fixed Charge} + \text{Meter Rent}$$

$$\text{Gross Amount} = 2104.53 + 15.00 + 10.00$$

$$\text{Gross Amount} = \text{Rs. } \mathbf{2,129.53}$$